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Practice Session 02 – Report

Cytoscape: importing, creating, and editing networks.

1. Nodes and edges in the networks

Network	Nodes	Edges
Marvel Universe Social network	6421	167112
Game of Thrones	84	216
University Classrooms	22	147

Table 1: Nodes and edges of the networks involved in this session.

2. Marvel Universe Social network

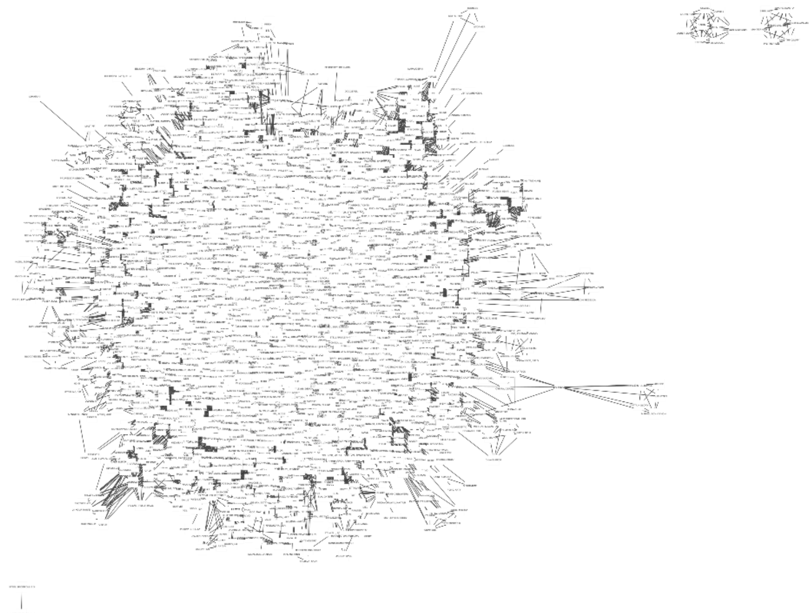

Figure 1: Marvel Universe Social network visualization with minimal style

Name	Degree	Name	Degree
CAPTAIN AMERICA	1905	VISION	1238
SPIDER-MAN/PETER PARKER	1737	INVISIBLE WOMAN/SUE	1236
IRON MAN/TONY STARK	1521	HAWK	1175
THING/BEN GRIMM	1416	WASP/JANET VAN DYNE	1091
MR. FANTASTIC/REED RICHARDS	1377	ANT-MAN/DR. HENRY J.	1082
WOLVERINE/LOGAN	1368	CYCLOPS/SCOTT SUMMER	1078
HUMAN TORCH/JOHNNY S	1361	SHE-HULK/JENNIFER WA	1071
SCARLET WITCH/WANDA	1322	STORM/ORORO MUNROE S	1070
THOR/DR. DONALD BLAK	1289	ANGEL/WARREN KENNETH	1070
BEAST/HENRY &HANK& P	1265	DR. STRANGE/STEPHEN	1065

Table 2: The twenty highest degree nodes in the Marvel Universe Social network

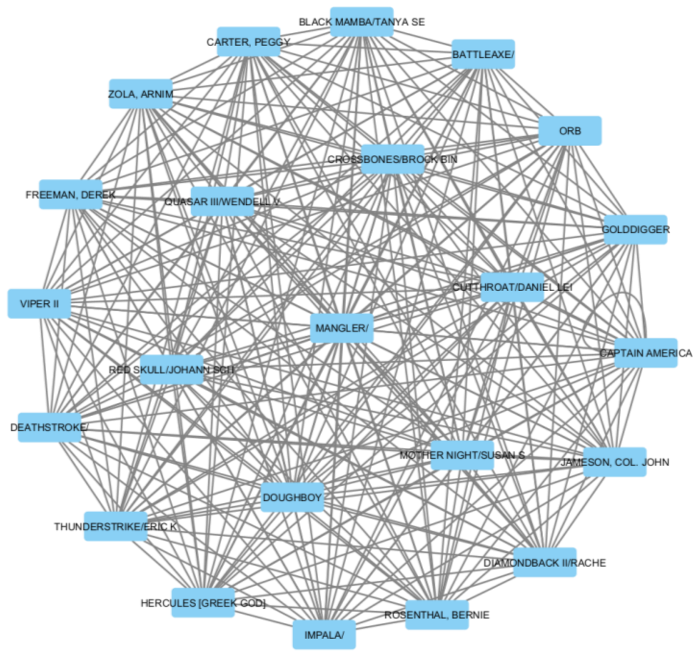

Figure 2: Neighbors of the node Deathstroke visualized

In the graph of Figure 2 there are 23 nodes with 254 edges in total. After applying the CoSE layout (the previous view was very messy) we can see that all the 22 nodes are neighbors of the Deathstroke node, but they are very interconnected between them almost like a clique. That makes sense in this network since if in a comic appear N characters, that will be represented by a clique of N nodes in the graph.

3. Game of Thrones network

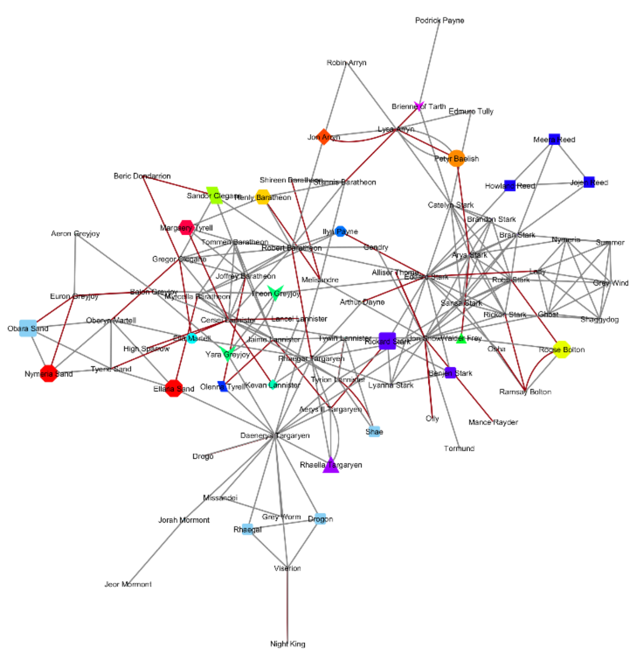

Figure 3: The Game of Thrones network visualized with styles

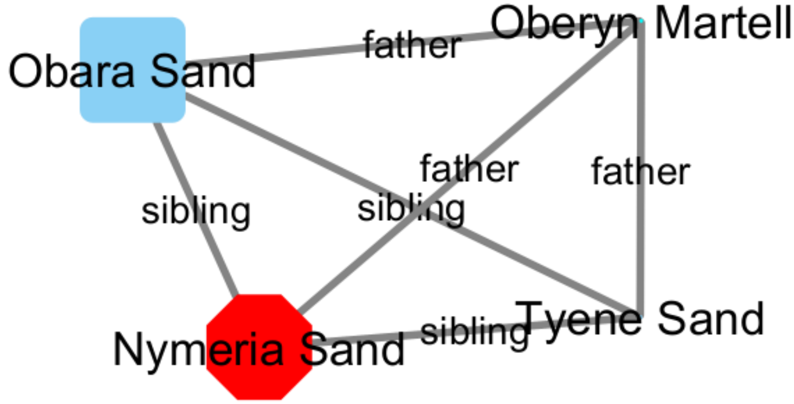

Figure 6: A clique in the Game of Thrones network of four nodes.

In Figure 4 we can see that Oberyn Martell is the father of three siblings: Obara Sand, Nymeria Sand and Tyene Sand.

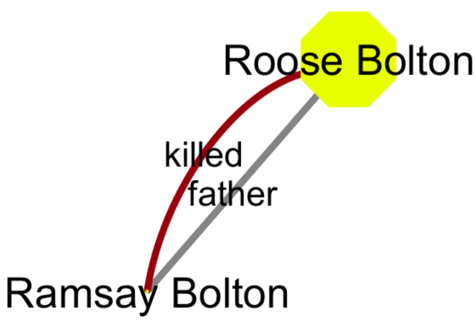

Figure 5: Multi-edge subgraph with two edges.

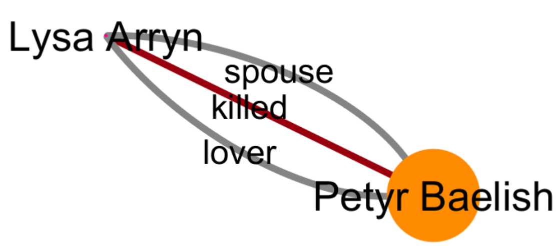

Figure 5: Multi edge subgraph with three edges.

In Figure 5 and Figure 6 we can see two different multi-edges. Once seen all the possible relationships, if we see a multi-edge it is very likely to be a plot-twist.

An observation that can be done with the “house of birth” of the graph is that, besides the house Targaryen, which is quite isolated because endogamy, all the other houses are quite interrelated. Those outer relationships are mainly because of a person killing another or because of marriage between houses. Within the houses (inner relationships) the main relationship is because of parenting or sibling.

4. University classrooms network

The network that can be seen in Figure 7 is composed of nodes that represent the classrooms in the university that I personally attended during the first trimester of last year. The edges between any pair of nodes mean that they belonged to the same subject. The tables that were used to build the network can be found in Annex: Tables of University Classrooms Network.

For instance, for a single subject that was taken in N different classrooms it would be represented as a clique of N nodes. From Figure 7 we can infer that there were 4 subjects in that trimester (namely

Operating Systems, Functional Program Design, Probability and Signals and Systems I) since we can clearly see that there are four different cliques (one at each cardinal direction) that are connected through some classrooms (e.g., 52.221, 52.321) because different subjects were taken in the same classroom. The coloring of the graph is yellow (BetweennessCentrality=0.0) when a classroom only is used in one subject and changes its color proportionally when it is related to more subjects.

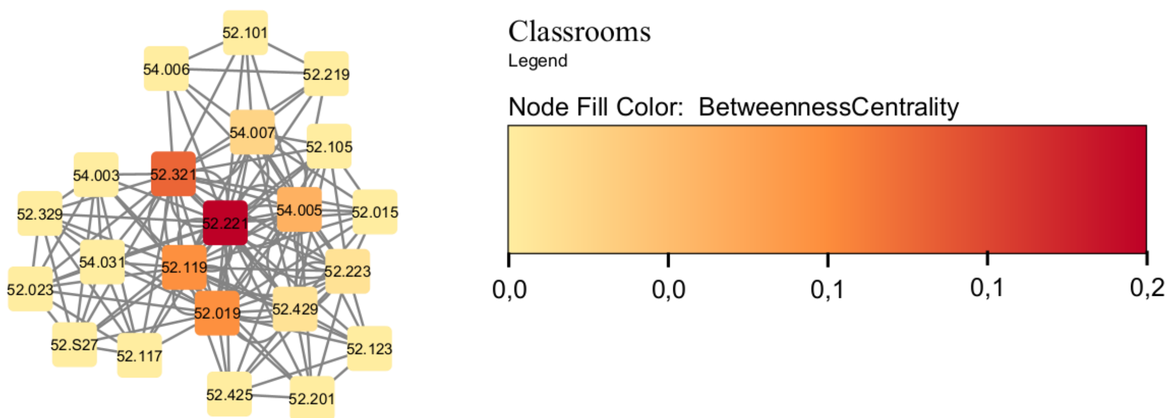

Figure 7: Classrooms network visualized.

I hereby declare that all of the text, tables, and figures in this report were produced by myself.

Annex: Tables of University Classrooms Network

id	classroom
0	52.015
1	52.019
2	52.023
3	52.101
4	52.105
5	52.117
6	52.119
7	52.123
8	52.201
9	52.219
10	52.221
11	52.223
12	52.321
13	52.329
14	52.425
15	52.429
16	54.003
17	54.005
18	54.006
19	54.007
20	54.031
21	52.S27

Table 3: Nodes of Classrooms Network

src	dest	subject
3	9	DFP
3	10	DFP
3	12	DFP
3	17	DFP
3	18	DFP
3	19	DFP
9	10	DFP
9	12	DFP
9	17	DFP
9	18	DFP
9	19	DFP
10	12	DFP
10	17	DFP
10	18	DFP
10	19	DFP

12	17	DFP
12	18	DFP
12	19	DFP
17	18	DFP
17	19	DFP
18	19	DFP
0	1	SiS1
0	4	SiS1
0	6	SiS1
0	10	SiS1
0	11	SiS1
0	12	SiS1
0	15	SiS1
0	17	SiS1
0	19	SiS1
1	4	SiS1
1	6	SiS1
1	10	SiS1
1	11	SiS1
1	12	SiS1
1	15	SiS1
1	17	SiS1
1	19	SiS1
4	6	SiS1
4	10	SiS1
4	11	SiS1
4	12	SiS1
4	15	SiS1
4	17	SiS1
4	19	SiS1
6	10	SiS1
6	11	SiS1
6	12	SiS1
6	15	SiS1
6	17	SiS1
6	19	SiS1
10	11	SiS1
10	12	SiS1
10	15	SiS1
10	17	SiS1
10	19	SiS1

11	12	SiS1
11	15	SiS1
11	17	SiS1
11	19	SiS1
12	15	SiS1
12	17	SiS1
12	19	SiS1
15	17	SiS1
15	19	SiS1
17	19	SiS1
1	6	POO
1	7	POO
1	8	POO
1	10	POO
1	11	POO
1	14	POO
1	15	POO
1	17	POO
6	7	POO
6	8	POO
6	10	POO
6	11	POO
6	14	POO
6	15	POO
6	17	POO
7	8	POO
7	10	POO
7	11	POO
7	14	POO
7	15	POO
7	17	POO
8	10	POO
8	11	POO
8	14	POO
8	15	POO
8	17	POO
10	11	POO
10	14	POO
10	15	POO
10	17	POO
11	14	POO
11	15	POO
11	17	POO

14	15	POO
14	17	POO
15	17	POO
1	2	P
1	5	P
1	6	P
1	10	P
1	12	P
1	13	P
1	16	P
1	20	P
1	21	P
2	5	P
2	6	P
2	10	P
2	12	P
2	13	P
2	16	P
2	20	P
2	21	P
5	6	P
5	10	P
5	12	P
5	13	P
5	16	P
5	20	P
5	21	P
6	10	P
6	12	P
6	13	P
6	16	P
6	20	P
6	21	P
10	12	P
10	13	P
10	16	P
10	20	P
10	21	P
12	13	P
12	16	P
12	20	P
12	21	P
13	16	P

13	20	P
13	21	P
16	20	P
16	21	P
20	21	P

Table 4: relations of Classrooms Network